Recursive and Iterative Inorder Traversal Comparison

1. Problem Statement: To develop a C++ program to implement recursive and iterative inorder traversal of the same binary tree and verify that both produce identical output.

2. Objectives:

• To implement recursive inorder traversal of a binary tree.

• To implement iterative inorder traversal using a stack.

• To compare the outputs of both traversals.

• To verify that both methods produce identical results.

3.Algorithm:

1. Start.

2. Create a binary tree with nodes containing data.

3. Initialize two empty vectors: recursiveResult and iterativeResult.

4. Perform recursive inorder traversal (Left → Root → Right) and store the output in recursiveResult.

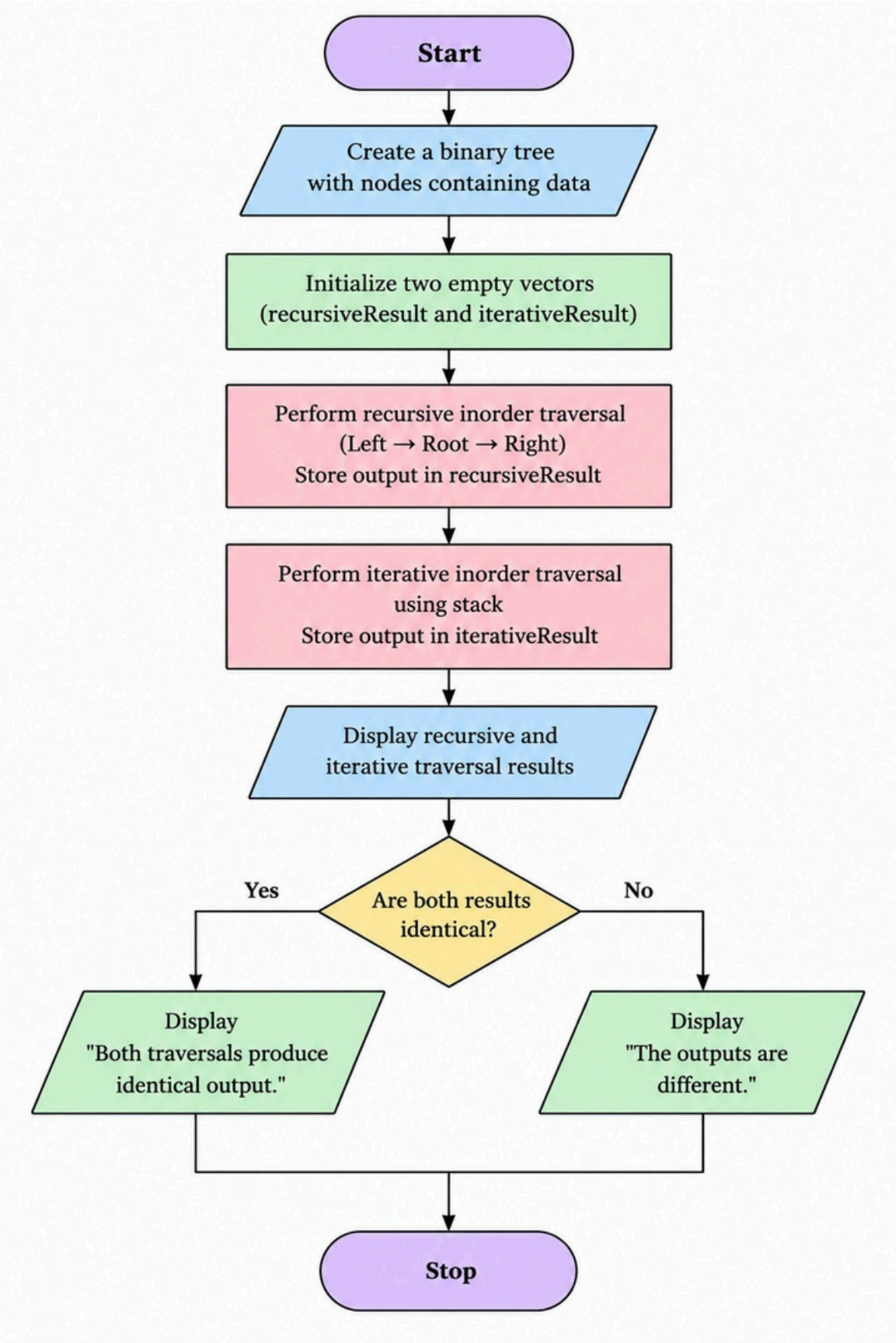
5. Perform iterative inorder traversal using a stack and store the output in iterativeResult.

6. Display both traversal results.

7. Compare both results. If they are identical, display "Both traversals produce identical output." Otherwise, display "The outputs are different."

8. Stop.

4. Flowchart


Start
Create a binary tree with nodes containing data
Initialize two empty vectors (recursiveResult and iterativeResult)
Perform recursive inorder traversal (Left → Root → Right) Store output in recursiveResult
Perform iterative inorder traversal using stack Store output in iterativeResult
Display recursive and iterative traversal results
Are both results identical?
Yes
No
Display "Both traversals produce identical output."
Display "The outputs are different."
Stop

5. Conclusion

The C++ program implements recursive and iterative inorder traversal of the same binary tree. Both methods follow the Left → Root → Right order and produce identical outputs. The comparison verifies that both traversal techniques work correctly.